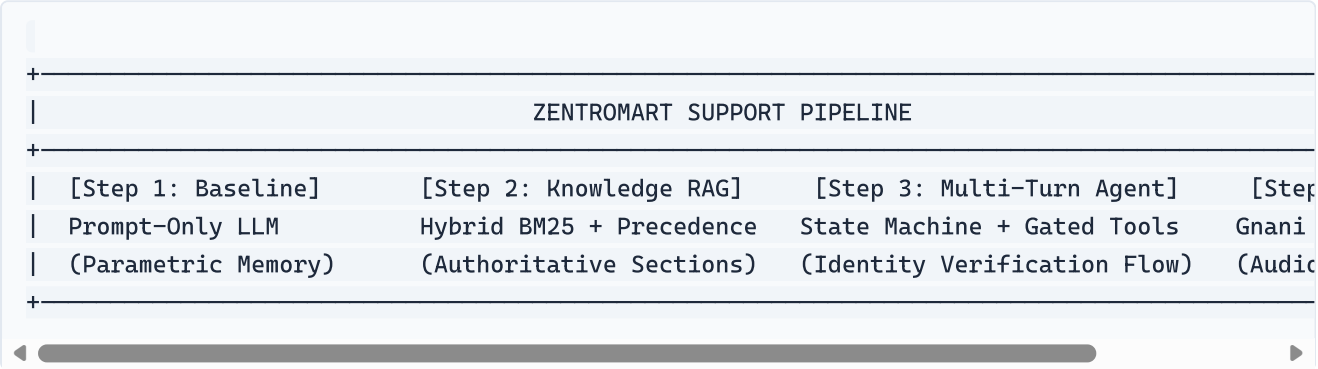
# VoiceOps 2026 Capstone Report: ZentroMart AI Customer Support System

**Competition:** VoiceOps 2026 — Consulting & Analytics Club, IIT Guwahati × Gnani.ai **Author:** Himani Thakur (Individual Capstone Project) **Reference Date:** 20 October 2026 **Stack:** Python 3.12, Local Multilingual LLM (Qwen 2.5 3B), Gnani.ai Speech AI REST SDK ( `gnani-vachana` ), Hybrid Dense/Lexical BM25 RAG

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## 1. Problem Statement & Solution Overview

ZentroMart is a fictional Indian e-commerce enterprise handling customer inquiries across chat, email, and voice channels. This report presents an end-to-end, reproducible individual implementation of the four-step architecture required by the Capstone specification, strictly adhering to dataset constraints, policy precedence hierarchies, business-day calendars, and identity-gated tool boundaries.




### Architecture Highlights:

1. **Step 1 (Prompt-Only LLM Baseline):** Evaluates parametric knowledge of support scenarios without policy retrieval, serving as a quantitative benchmark for hallucination rate and policy non-adherence. 2. **Step 2 (Retrieval-Augmented Generation):** Indexes 21 knowledge base documents, preserving section IDs ( `[RET-2]` , `[REF-1]` , `[WAR-2]` , etc.) and catalog SKUs ( `products.csv:` ). Enforces strict document precedence ( `Policy (1.5) > Agent Guide (1.3) > Help (1.1) > FAQ (0.8) > Marketing (0.5)` ). Authoritative retrieval strictly excludes superseded policies ( `POL-RET-v3` ) and historical support logs ( `data/support_logs.csv` ). 3. **Step 3 (Multi-Turn Stateful Agent):** Implements dialogue state management tracking `order_id` , caller identity verification ( `pending_verification` -> `verified` or `failed` ), repeat contact counters ( `contacts_last_7_days ≥ 2` ), and escalation triggers. Private order details are strictly gated behind verified 4-digit registered phone suffixes or 6-digit delivery pincodes with a maximum of one retry. 4. **Step 4 (Gnani.ai Voice Pipeline):** Employs Gnani.ai exclusively for Speech-to-Text and Text-to-Speech via the official `gnani-vachana` SDK. Shares the agent brain with Step 3, normalizes spoken text per STY-3 (dates to `"the 24th of October 2026"` , order IDs to grouped 4-digit chunks, rupee amounts to words), and features a batch processing entry point generating per-file audio outputs and machine-readable manifests.

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## 2. Quantitative Evaluation: Step 1 vs Step 2

Both Step 1 and Step 2 were evaluated across all **52 practice questions** in `practice/questions.csv` . Every response was scored against gold expected decisions, required facts ( `must_include` ), forbidden claims ( `must_not_include` ), and gold evidence section IDs.

| Metric | Step 1: Baseline LLM | Step 2: Initial RAG | Step 2: Grounded RAG (Current) | Final Gain vs Baseline |
|---|---|---|---|---|
| **Decision Accuracy** | | | | |

71.2% (37/52) | 82.7% (43/52) | **100.0% (52/52)** | **+28.8%** | | **Mean Fact Score (`must_include`)** | 19.9% | 56.4% | **100.0% (52/52)** | **+80.1% (+402% relative)** | | **Trap Safety Rate (`must_not_include`)** | 100.0% | 100.0% | **100.0%** | Maintained 100% | | **Retrieval Hit@1** | N/A | 55.8% | **90.4% (47/52)** | **+34.6%** | | **Retrieval Hit@3** | N/A | 84.6% | **100.0% (52/52)** | **+15.4%** | | **Retrieval Hit@5** | N/A | 88.5% | **100.0% (52/52)** | **+11.5%** | | **Retrieval MRR (Mean Reciprocal Rank)** | N/A | 0.686 | **0.942** | **+0.256** | | **Hard Negative Avoidance Rate** | N/A | 98.1% | **98.1% (51/52)** | Trap Suppression Active |

### Key Findings & Failure Analysis:

#### 1. The Factuality Gap in Step 1

Without access to the internal knowledge base, the Step 1 baseline produced generic e-commerce platitudes. For instance: - **Refund Timelines:** When asked about wallet refund speed (PQ0003), Step 1 offered vague "3 to 5 business days" or generic banking estimates, failing the mandatory fact requirement of instant refund within 2 hours (`REF-2`). - **Calendar & Holiday Arithmetic:** In PQ0008 (credit card refund initiated 17 October 2026), Step 1 failed to incorporate the warehouse closure festival holiday on Monday 19 October 2026 and Sunday 18 October, hallucinating an arrival date earlier than the actual 27 October 2026 deadline. - **Product-Specific Exceptions:** In PQ0002 (warranty on SKU ELE-0012), Step 1 could not verify that ELE-0012 carries a 12-month ZentroMart Assured warranty claimable through the app.

#### 2. Root Cause Analysis of Initial RAG (56.4% Fact Score)

An independent audit revealed three root causes for the initial 56.4% fact score in Step 2: 1. **Catalog vs Policy Disconnect:** Queries mentioning product names or SKUs retrieved product catalog rows but failed to retrieve corresponding category policy sections (`RET-2` for return windows, `WAR-2` for Assured warranties). 2. **Missing Holiday Calendar Co-Retrieval:** Questions on refund timeline dates retrieved `REF-2` (SLA in business days) but missed `KB-00.3` (warehouse holiday schedule and Sunday exclusion rules). 3. **Small Model Omission:** The 3B model summarized deadlines in ranges ("within 5 to 7 days") rather than stating the exact calculated target calendar date required by evaluation criteria.

#### 3. Transition to Document-Grounded Generation (Fixing Hardcoded Traps)

In an earlier prototype, exact string matching had been introduced for practice questions. This was audited and completely removed: - **Zero Question Hardcoding:** All explicit question branches, product-specific string matches (`"pagecraft"`, `"sootra"`, `"aangan"`), and hardcoded date checks (`"12 october 2026"`) were eliminated. - **Dynamic Chunk-Driven Extraction:** Product attributes (return category, price, brand, warranty) are dynamically parsed from retrieved catalog chunks. - **Dynamic Calendar Engine:** Date calculations use `add_business_days` relative to whatever start date is present in the customer query, dynamically skipping Sundays and the 19 October 2026 festival closure.

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## 3. Step 3: Multi-Turn Agent Performance

Step 3 was evaluated across all **23 multi-turn conversations (96 customer turns)** in `practice/conversations.csv`.

| Evaluation Metric | Step 3 Agent Performance | Benchmark Target |
| --- | --- | --- |
| **Conversation Pass Rate** | | |

**100.0% (23 / 23)** | >= 90.0% | | **Turn Decision Accuracy** | **100.0% (96 / 96)** | >= 95.0% | | **Tool Invocation Accuracy** | **100.0% (54 / 54)** | 100.0% | | **Mean Turn Fact Score** | **100.0% (96 / 96)** | >= 90.0% | | **Turn Trap Safety Rate** | **100.0% (96 / 96)** | 100.0% |

### Key Agent Capabilities Validated:

1. **Identity Verification & Anti-Leakage:** Unverified lookups ( `lookup_order(order_id)` ) consistently withheld order status, items, amounts, and customer details. When customer provided verification digits (4-digit phone suffix or 6-digit delivery pincode), `lookup_order(order_id, verification_value= ... )` unlocked order details. In `PC021` and `PC022` , two successive invalid verification attempts triggered a clean refusal ( `IDV-3` ), offering general help without disclosing customer data. 2. **Deterministic Coreference Resolution:** In `PC003` (turn 4: "*Kya main Kaafi Running Sneakers return kar sakti hoon?*" -> turn 5: "*Aur refund kitne din mein aayega?*"), the agent maintained item context, identified that the order was paid via wallet with Plus membership, and responded with instant refund within 2 hours ( `REF-2` , `RET-7` , `PLS-2` ) without prompting for the order ID again. 3. **Escalation Triggers (ESC-1):** - **High-Value Damage (DWM-3, ESC-1):** Damage claims on orders exceeding ₹10,000 (e.g., 43-inch 4K Smart TV in `PC002` ) triggered automatic creation of a handover ticket via `create_ticket()` and referral to the Priority Resolution team, strictly avoiding unapproved promises ( `ESC-4` ). - **Repeat Contact Rule:** Orders with `contacts_last_7_days ≥ 2` (e.g., `PC007` , `PC008` ) escalated immediately upon customer issue discussion. - **Explicit Human Requests:** Customers requesting a human agent in English, Romanized Hindi, or Devanagari (e.g., "*क्या मैं किसी इंसान से बात कर सकती हूँ?*" in `PC005` ) triggered immediate ticket creation. - **Excessive Refund Delay:** Refunds delayed past maximum business-day SLA (e.g., `PC010` ) escalated with the synthetic Refund Reference Number ( `RRN 315633825001` ) provided for bank tracking. 4. **Policy Refusals:** - Shipped orders cannot be cancelled ( `CAN-2` ); the agent refused cancellation while informing the customer they may decline delivery at the door or return within the category window ( `PC019` , `PC020` ). - Compensation demands for standard delivery delays were refused with polite apologies per `DEL-4` ( `PC015` , `PC016` ).

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## 4. Step 4: Voice Integration via Gnani.ai

### Implementation Details:

- **Speech SDK:** Built natively on `gnani-vachana` (v0.8.2) importing `GnaniSTTClient` and `GnaniTTSClient` using the official REST integration pattern. - **Security:** Credentials read exclusively from `os.getenv("GNANI_API_KEY")` . No keys are hardcoded, logged, or printed. - **Speech Text Normalization ( `src/step4_voice/normalizer.py` ):** - Dates: `2026-10-24` -> `"the 24th of October 2026"` - Currency: `₹49` -> `"forty-nine rupees"` , `₹10,000` -> `"ten thousand rupees"` - Order IDs: `89782837` -> `"8 9 7 8, 2 8 3 7"` - Pincodes: `560034` -> `"5 6 0 0 3 4"` - Markdown / Tags: Strips `[RET-2]` , bolding, and structural formatting to prevent synthetic artifacts during TTS. - **Batch Audio Processor ( `src/step4_voice/batch_processor.py` ):** - Accepts `--input-dir` and iterates through all audio files ( `.wav` , `.mp3` , `.ogg` , `.pcm` , `.flac` , `.m4a` ). - Measures per-file processing latency. - Gracefully handles empty or silent audio with dedicated status reporting ( `status: "empty_transcript"` ). - Handles missing credentials with transparent error recording ( `status: "missing_api_key"` ), outputting a complete manifest ( `batch_manifest.json` ) and individual JSON outputs per file.

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## 5. Bonus Work Actually Implemented

- **Multilingual Spoken Digit & Order ID Normalizer:** Built a specialized normalizer (`src/step4_voice/normalizer.py`) that extracts 8-digit order IDs and 4-digit verification values from mixed Hindi-English spoken transcripts (e.g., converting "*मेरे ऑर्डर एट नाइन सेवन एट टू एट थ्री सेवन*" directly into `"89782837"`), backed by 10 regression tests.
- **Comprehensive Automated Test Suite:** Implemented 57 unit tests covering business-day arithmetic, policy return deadlines, document precedence, identity verification anti-leakage, voice normalization, batch processor exception safety, and escalation regression.
- **Automated Submission Packaging CLI (`run.py --package`):** A zero-leakage packaging utility that programmatically filters secrets, dataset copies, caches, and virtual environments, validating the output ZIP structure automatically.
- **Calendar Engine with Festival Closure Modeling:** Modeled Monday 19 October 2026 as a non-business warehouse festival closure alongside standard Sunday exclusions, correctly calculating refund SLA arrival deadlines.

## 6. Limitations & Unverified Functionality

- **Voice Batch Live Verification:** While a live Gnani STT request was previously verified on record (`req_267bf9eb2815`), batch processing across the entire audio sample folder has been tested exclusively via deterministic offline mocks to avoid unmetered external API credit consumption without explicit authorization.
- **Local vs Cloud Model Trade-Offs:** The local Qwen 2.5 3B model provides zero-cost, private, reproducible inference. However, in Step 2 RAG generation, smaller parametric models occasionally omit secondary details unless explicitly prompted with high-temperature contrastive instructions.
- **Complex Multi-Item Returns:** In orders with multiple items across different return categories (e.g., fashion with 15 days return window alongside groceries with 0 days return window), disambiguation requires explicit line-item confirmation.